The Foreign Policy Impact of Trade-Based Status Gains: When China Overtakes the US as Top Export Destination

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Abstract

How do trade-based status gains affect UNGA voting convergence toward China? Existing IR research emphasizes status anxiety and status seeking, but pays less attention to upward status shifts. I argue that rank reversals in trade hierarchies, especially when a rising power displaces the hegemon, create a salient status cue beyond gradual trade growth. The salience shock can be immediate, but diplomatic responses may accumulate gradually as policymakers revise positions under institutional inertia and reputational constraints. I test this claim in Brazil, where China overtook the United States as Brazil's largest export destination in 2009, using Synthetic Difference-in-Differences on UNGA ideal-point distance. The estimate implies a statistically significant 41% average post-2009 reduction in Brazil's distance to China relative to the synthetic counterfactual ($p = 0.032$). Brazilian newspaper evidence is consistent with a post-2009 increase in China-related salience, particularly trade-focused coverage, but it does not identify the full salience-to-vote mechanism. I then examine a scope-conditioned cross-country sample of countries for which the United States was the relevant benchmark or prior top export partner. A switching-treatment fect IFE specification provides suggestive scope evidence consistent with a similar pattern within this sample. These results indicate that discrete status milestones, not only continuous trade deepening, can be associated with selective UNGA voting convergence toward China in hegemonic-benchmark settings.

Word count: 6,920

1 Introduction

"China overtakes US as Brazil's top trade partner." That headline captured how Brazilian newspapers framed a specific 2009 event: China had become Brazil's largest export destination (Rodrigues 2009). The phrase is useful because it records the public language of the moment. Empirically, however, the treatment in this paper is narrower and consistent across designs: China becomes the largest export destination.

The paper asks whether this kind of rank threshold matters for one observable dimension of foreign-policy behavior: UNGA voting convergence toward China. Existing work usually treats trade exposure as continuous. As exports, imports, loans, or investment rise, foreign-policy alignment is expected to shift gradually through material leverage, changing interests, or domestic pressure (Kastner and Pearson 2021; Yan and Zhou 2021; Morgan 2019; Flores-Macías and Kreps 2013). That logic misses a separate possibility. A number-one rank can make an already growing relationship politically legible. When a rising power crosses the top-rank threshold, especially by displacing a prominent incumbent, the event can become a focal cue for media, elites, and policymakers.

I separate three empirical claims. The main claim is a reduced-form Brazil claim: after China became Brazil's largest export destination in 2009, Brazil's UNGA ideal-point distance to China fell relative to a synthetic counterfactual. The second claim is a Brazilian media-salience claim: Folha de S.Paulo coverage shows that China and China-Brazil trade became more visible around the rank reversal. This evidence supports the attention step of the theory, not the full causal path from media attention to UNGA votes. The third claim is a cross-country scope claim: in a scope-conditioned panel, similar top-export-destination transitions are associated with smaller UNGA distance to China, but this evidence is a scope probe rather than a second Brazil-style identification design.

Brazil is the strongest case for the theory. China displaced the United States, a politically salient incumbent and hegemonic benchmark; the rank reversal was publicly visible; and the Brazilian press allows the attention implication to be measured. I estimate the Brazilian effect using synthetic difference-in-differences (SDiD) on annual UNGA ideal-point distance to China. The estimate is an average post-2009 gap, not a one-year discontinuity. Relative to Brazil's pre-treatment mean distance, the estimated reduction is 40.8 percent.

The paper also narrows the substantive interpretation. The evidence shows selective UNGA voting convergence toward China after a salient trade-rank reversal. Outcome diagnostics indicate that the result is not an artifact of a single absolute-distance metric, but resolution-level evidence is issue-sensitive and points especially to areas where Brazil and China had room to converge, including human rights. The contribution is therefore about trade-rank thresholds as politically usable status cues, not about a general replacement of Brazil's foreign-policy orientation.

2 Theory and hypotheses

A rank reversal differs from smooth trade growth because it changes a category. Actors can observe trade shares, but they also use coarse labels such as largest export market, number-one partner, or top trade partner. These categories simplify complex economic relationships and help journalists, politicians, and organized interests coordinate attention (Bordalo, Gennaioli, and Shleifer 2022; Conlon 2025; Enke 2024; Graeber, Roth, and Sammon 2025). The theoretical claim is not that trade volumes are irrelevant. It is that crossing the number-one threshold can create a politically salient cue beyond the information contained in continuous trade exposure.

The cue should matter most when the displaced incumbent is politically prominent. A Chinese rank reversal over a minor incumbent may be a commercial fact with little political resonance. A Chinese rank reversal over the United States is different. It can be read as a visible status loss for the hegemon and as a public marker of China's rise. Brazil falls squarely inside this high-salience scope condition: China displaced the United States in the export-destination hierarchy in 2009, and contemporary coverage frequently described the event in broader top-partner language.

The mechanism has two steps. First, the rank reversal should increase attention. Media coverage should make China and the bilateral trade relationship more visible after the threshold is crossed. Second, the salience cue may enter policy deliberation through diplomatic, reputational, and coordination channels. This second step can be gradual because UNGA positions are embedded in prior votes, bureaucratic routines, coalition commitments, and reputational constraints. The Brazilian SDiD design estimates the combined reduced-form effect of the rank reversal on UNGA distance to China; it does not separately identify each channel.

The main expectation is therefore: when China becomes a country's largest export destination, that country's UNGA ideal-point distance to China should decrease relative to its counterfactual path. This expectation is tested most directly in Brazil. A second expectation is a scope condition rather than a confirmed moderation result: the effect should be more politically salient when China displaces a long-entrenched U.S. incumbent. The Brazilian case satisfies this condition, while the cross-country panel treats U.S.-displacement heterogeneity as exploratory because incumbent identity is not randomly assigned and subgroup support is limited. A third expectation is that media coverage of China and China-Brazil trade should increase around the rank reversal. A fourth expectation is temporal: if the effect depends on the continuing salience of the top-rank cue, it should weaken when China loses that position, although exit diagnostics alone cannot distinguish salience from reversible interest-based mechanisms.

3 Data and design

Empirically, the treatment is China becoming a country's largest export destination. I use "top trade partner" only as shorthand for the publicly salient Brazilian event when discussing contemporary framing. The Brazil SDiD and cross-country panel define treatment using largest-export-destination status.

The primary outcome is absolute UNGA ideal-point distance to China, based on the annual ideal-point estimates of Bailey, Strezhnev, and Voeten (2017). Lower values indicate convergence toward China. This measure is preferable to raw annual agreement shares for the main design because it is less sensitive to changes in the UNGA agenda over time. I use resolution-level agreement diagnostics only to interpret where convergence appears strongest.

For Brazil, the treatment indicator equals one from 2009 onward, the first year China became Brazil's largest export destination. SDiD estimates Brazil's average post-2009 gap relative to a weighted synthetic counterfactual, using pre-treatment information to construct unit and time weights (Arkhangelsky et al. 2021). The design is reduced-form. Its credibility rests on pre-treatment fit, rank-versus-volume timing tests, donor-pool sensitivity, and outcome robustness, not on identifying every mechanism linking salience to votes.

The SDiD panel combines UNGA ideal points with trade, macroeconomic, power, distance, political-institution, and trade-agreement covariates. The complete panel covers 96 countries and 19 Brazil years for the main SDiD window (1997-2015). Trade data come from the International Trade and Production Database for Estimation (Borchert et al. 2021). The ITPD-E `trade` variable is measured in millions of current US dollars, and export-destination ranks are computed from exporter-importer flows.

For the cross-country scope probe, treatment equals one in years when China is currently the country's largest export destination and returns to zero if China loses that position. This switching structure differs from absorbing-treatment staggered DiD. The main panel estimator is the counterfactual `fect` estimator with interactive fixed effects (Liu, Wang, and Xu 2024), using log GDP per capita and V-Dem freedom of expression as covariates. Absorbing-treatment estimates are reported in the appendix because they estimate a different, persistent-treatment diagnostic quantity.

4 Brazil SDiD evidence

The Brazilian analysis estimates the average post-2009 reduced-form effect of China's entry into the top export-destination position on Brazil's UNGA ideal-point distance to China. The design does not identify each mechanism linking salience to votes. It asks whether Brazil moved closer to China than a synthetic counterfactual after the 2009 rank reversal.

Figure 1 shows why the treatment is not just smooth export growth. China's export share rose before 2009, but China became the top-ranked export destination only in 2009. The signed margin over the relevant competitor also changes sign at that threshold.

Figure 2 presents the main SDiD fit. The pre-treatment fit is close enough to make the post-treatment comparison informative, and the post-2009 gap is negative, indicating convergence toward China. The point estimate is -0.26 ideal-point units, with placebo-based SE 0.12 and p-value 0.032. Relative to Brazil's pre-treatment mean distance (0.647), this corresponds to a 40.8 percent reduction.

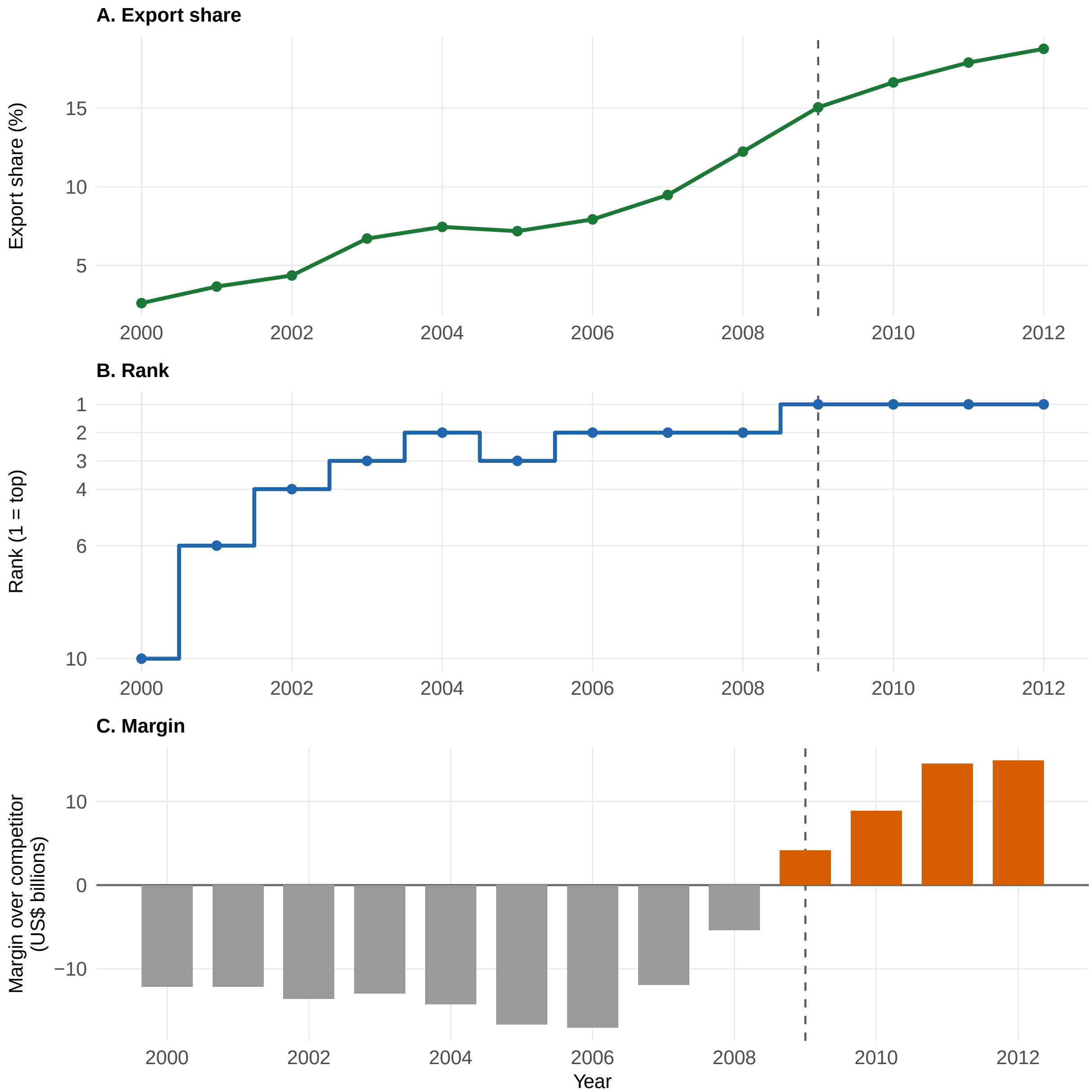

Figure 1: Brazil rank-versus-volume diagnostic for China's export-destination status. Panel A shows China's share of Brazilian exports. Panel B shows China's ordinal export-destination rank, with 1 as the top rank. Panel C shows China's signed export-value margin over the relevant competitor: the destination China had to surpass before 2009 and the second-ranked destination after China became first. Orange bars indicate positive margins and gray bars indicate negative margins. The dashed line marks 2009, when China became Brazil's largest export destination. Note: Unit = export share in percent; ordinal rank; export-value margin in current USD billions based on ITPD-E trade values. Time window = 2000-2012 annual Brazil series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

The estimate should not be read as a fall of 0.26 points in 2009 itself; it summarizes the average post-treatment gap.

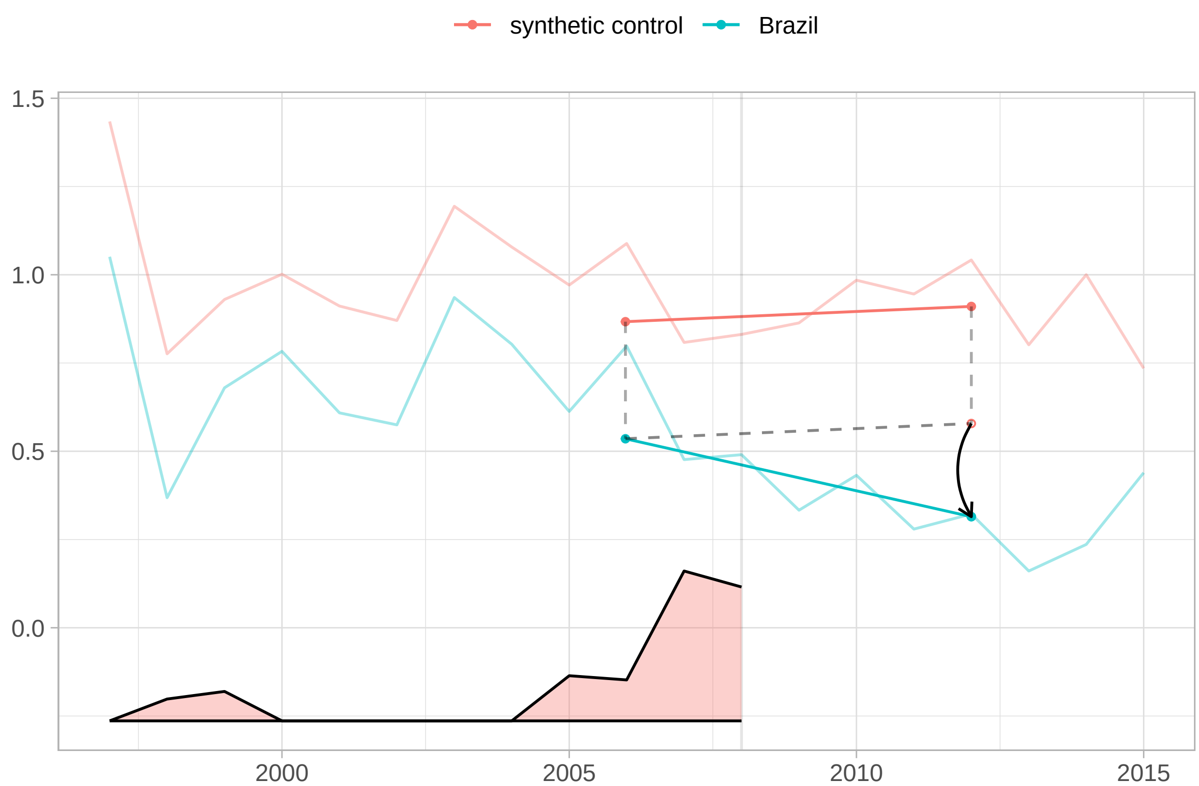

Figure 2: SDiD fit for Brazil and synthetic control with time weights. Lower values indicate convergence toward China. Uncertainty is reported in Table 1, not plotted here. The post-treatment gap is interpreted as an average post-2009 reduced-form effect, not as a one-year discontinuity. Note: Unit = absolute UNGA ideal-point distance to China (points). Standard error = placebo-based SE and p-values reported in the table. Time window = Brazil annual SDiD series with treatment in 2009. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

The rank interpretation requires more than showing that trade with China was increasing. Brazil's exports to China grew before 2009, including years in which China rose in the export hierarchy but did not become number one. The timing tests in Table 1 therefore ask whether rapid trade growth and lower-rank promotions generate the same estimated UNGA convergence as the top-rank reversal. They do not. The 2003 and 2005 rows are timing falsifications for the continuous-growth alternative, not equivalence tests. The 2012 row probes whether the result is driven by a later break after China had already become the top destination.

Table 1: Brazil SDiD estimate and rank-versus-volume timing tests.

Timing year	Test role	China rank	Rank-1 reversal	Export share (%)	Margin (USD bn)	ATT	Placebo SE	p-value
2003	Growth / lower-rank promotion	3	No	6.7	-12.9	-0.117	0.188	0.534
2005	Rapid growth without rank 1	3	No	7.2	-16.7	-0.119	0.135	0.379
2009	Actual rank-1 reversal	1	Yes	15.1	4.1	-0.264	0.123	0.032
2012	Later-break falsification	1	No	18.8	14.9	0.037	0.139	0.792

Note:

Unit = ATT in absolute UNGA ideal-point distance to China; lower values indicate convergence toward China; export margins are current USD billions based on ITPD-E trade values. Standard error = normal approximation using placebo-based SE. Time window = main model uses treatment in 2009; 2003 and 2005 timing tests truncate the sample before 2009; 2012 is a later-break falsification. Treatment definition = China becomes Brazil's largest export destination in 2009; placebo rows redefine treatment at the pseudo-treatment year. The 2003 and 2005 rows are timing falsifications for rapid export growth without rank-1 status, not equivalence tests.

Outcome robustness broadens the measurement base but narrows the substantive claim. Table 2 shows that the main result is not an artifact of the absolute-distance metric: the China-minus-US relative-distance contrast also points toward convergence with China. Agreement-based outcomes and resolution-level diagnostics are more sensitive to agenda composition, so I treat them as interpretive diagnostics rather than alternative primary estimands. The resolution-level evidence points to selective convergence, especially in human-rights votes, not a uniform shift across the UNGA agenda.

Table 2: Brazil outcome robustness and issue-scope diagnostics.

Outcome / diagnostic	Unit	Estimate	Causal status	Interpretation
Absolute Brazil-China ideal-point distance	country-year SDiD outcome	-0.264 (SE 0.123)	Brazil SDiD reduced-form estimate with placebo SE	Primary reduced-form estimate
China-minus-US ideal-point distance	country-year SDiD outcome	-0.521	alternative country-year SDiD robustness; point estimate only	Direction consistent with convergence toward China relative to the United States
Absolute ideal-point distance to the United States	country-year SDiD outcome	0.305	secondary country-year SDiD diagnostic; point estimate only	Agenda-sensitive diagnostic; point estimate only
Annual vote agreement with China	country-year SDiD outcome	0.078	agenda-sensitive country-year diagnostic; point estimate only	Agenda-sensitive diagnostic; point estimate only
Annual agreement with China minus agreement with the United States	country-year SDiD outcome	0.066	agenda-sensitive country-year diagnostic; point estimate only	Agenda-sensitive diagnostic; point estimate only
Human rights identical-vote change	resolution-level descriptive diagnostic	+7.5 p.p.	descriptive pre/post diagnostic, not SDiD	Larger descriptive convergence where Brazil had room to move
Non-human-rights identical-vote change	resolution-level descriptive diagnostic	+2.7 p.p.	descriptive pre/post diagnostic, not SDiD	Smaller descriptive change outside human rights

Note:
Unit = country-year SDiD estimates and resolution-level percentage-point changes. Standard error = placebo-based SE only for the primary absolute-distance outcome; alternative SDiD outcomes are point estimates. Time window = Brazil country-year outcomes use the SDiD panel; resolution-level diagnostics compare 2005-2008 with 2009-2012. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009. Agreement and resolution-level measures are agenda-sensitive diagnostics. They narrow the substantive interpretation to selective UNGA voting convergence toward China.

The Brazilian design cannot eliminate every possible contemporaneous political development in 2009. Its credibility comes from the combination of close pre-treatment fit, explicit covariates for trade exposure and macroeconomic stress, timing tests during earlier China trade growth, and donor-pool sensitivity checks reported in the appendix. The estimate should therefore be read as strong reduced-form evidence for the Brazilian rank-reversal episode, not as a direct test of every mechanism linking salience to votes.

5 Brazilian media salience

The SDiD estimate is reduced-form. To assess whether the rank reversal became publicly salient, I examine Folha de S.Paulo coverage of China from 2000 to 2014. This evidence measures the attention step of the theory: whether China and China-Brazil trade became more visible in public discourse after the rank reversal.

The corpus contains headlines and related news items mentioning China. I classify China-related coverage into broad categories and focus on whether bilateral trade coverage grows after 2009. Figure 3 shows that the post-2009 increase is concentrated in China-Brazil trade coverage rather than in all China-related topics equally. This pattern is consistent with a rank-salience interpretation: the public cue was not just China in general, but China's new commercial status in Brazil.

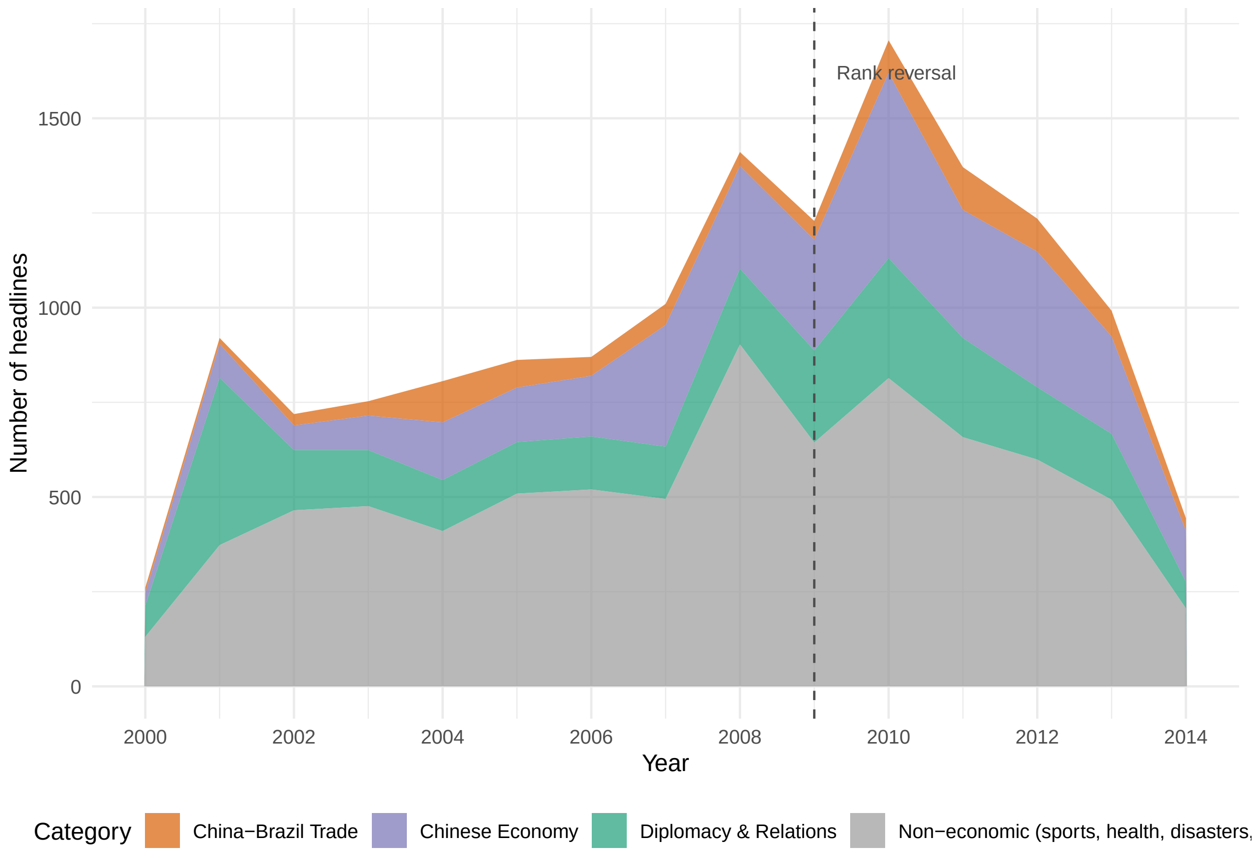

Figure 3: Brazilian media salience: disaggregated China headlines in Folha de S.Paulo by category. The sustained post-2009 increase is concentrated in trade-related coverage. This figure is descriptive salience evidence and does not identify the effect of media attention on UNGA votes. Note: Unit = count of headlines per category per year. Time window = 2000-2014 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

This evidence identifies the first step in the proposed mechanism. It shows that the rank reversal became more visible in Brazilian public discourse and that the increase was concentrated in trade-related coverage. It does not prove that media attention caused UNGA votes to change. The policy-response channel remains a theoretically motivated interpretation of the reduced-form SDiD estimate, supported by timing and content, but not separately identified.

6 Cross-country scope probe

The panel analysis should not be read as a second Brazil. It estimates a broader and weaker estimand: the average association between China entering and holding the top export-destination position across heterogeneous cases in a U.S.-benchmark sample. Its purpose is to probe scope and reduce the concern that the Brazil result is purely idiosyncratic. It does not measure media salience in each country and does not identify the full mechanism.

The preferred scope-probe specification is the switching-treatment `fect` IFE model with log GDP per capita and press freedom. The estimate is -0.073 (bootstrap SE = 0.033, 95 percent CI [-0.138, -0.008], p = 0.027). It corresponds to about 12.6 percent of the pre-treatment mean distance to China, or 0.103 standard deviations of the outcome. The estimate is in the theoretically expected negative direction and statistically distinguishable from zero at the 5 percent level. Table 3 summarizes the specification.

Table 3: Cross-country scope-probe estimate for China top export-destination status.

Quantity	Value
Estimator	fect IFE with covariates
Treatment type	Switching
ATT	-0.073
Bootstrap SE	0.033
95% CI	[-0.138, -0.008]
p-value	0.027
N (country-years)	2627
Treated units	29
Control units	58
Covariates	log GDP pc, V-Dem free expression
Latent factors	r* = 1 (cross-validation)
Panel window	1990–2020

Note:

Unit = ATT in absolute UNGA ideal-point distance to China; lower values indicate convergence toward China. Standard error = bootstrap with 10,000 replications. Cluster = country level. Time window = 1990–2020 covariate complete-case switching panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination within the scope-conditioned panel; turns off if China loses the top position (switching treatment). This estimate is a scope probe. Callaway-Sant'Anna absorbing-treatment estimates are appendix sensitivity checks because they estimate a persistent-treatment subset rather than the same switching estimand.

Figure 4 shows the entry-aligned dynamic pattern for the covariate-adjusted `fect` IFE specification. The pre-entry estimates are diagnostics for model fit, and the post-entry path indicates whether the panel pattern is consistent with the Brazil result. The figure is not mechanism evidence.

The pooled panel does not test the hegemonic-replacement scope condition directly because it averages across treatment spells with different displaced incumbents and salience levels. I therefore treat U.S.-displacement heterogeneity as exploratory rather than confirmatory evidence for H2. In the available spell-level comparison, U.S.-displacement cases do not produce a clearly larger estimated effect than non-U.S.-displacement cases. This prevents a strong cross-country claim that H2 has been confirmed, while leaving the Brazilian case as the high-salience setting for the main reduced-form design.

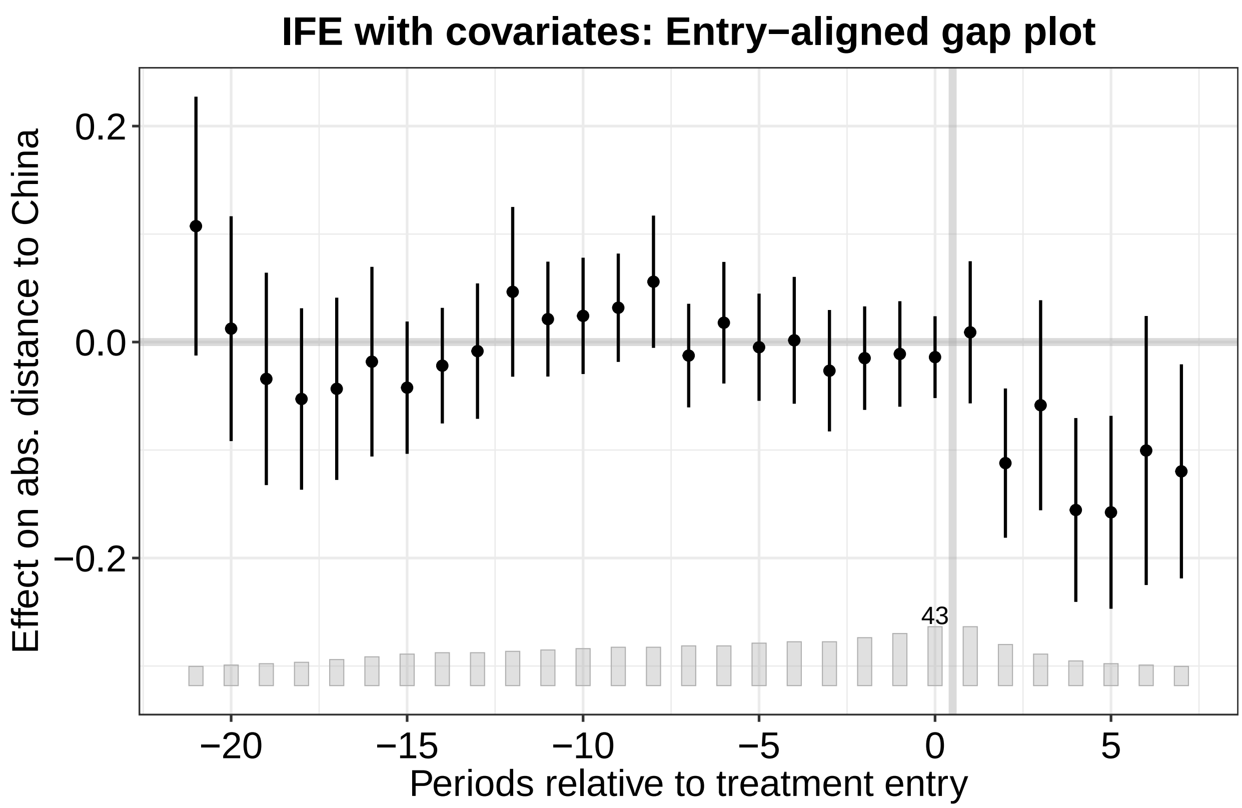

Figure 4: Cross-country dynamic scope-probe figure from the covariate-adjusted fect IFE model. Effects are aligned by treatment entry. This figure probes whether the top-rank pattern travels beyond Brazil; it does not identify the salience mechanism. Note: Unit = effect on absolute UNGA ideal-point distance to China (points). Standard error = bootstrap confidence intervals from fect target. Time window = 1990–2020 covariate complete-case switching panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination within the scope-conditioned panel; turns off if China loses the top position (switching treatment).

The cross-country estimates should be interpreted cautiously. The panel does not measure media salience in each country, treatment spells differ in incumbent identity and political visibility, and exit/carryover diagnostics cannot by themselves distinguish salience from reversible interest-based mechanisms. The panel therefore probes whether the top-rank pattern travels beyond Brazil; it does not identify the full mechanism.

7 Conclusion

The evidence indicates that rank thresholds in trade hierarchies can matter for one observable dimension of foreign-policy behavior: UNGA voting convergence toward China. The strongest evidence comes from Brazil's 2009 rank-reversal episode, where SDiD estimates show a post-treatment reduction in distance to China and media evidence shows a contemporaneous rise in China-Brazil trade salience. Cross-country evidence is consistent with a broader top-rank pattern, but it remains a scope probe.

The findings speak to debates about whether status gains have observable behavioral consequences by showing that a specific economic status cue can be associated with measurable voting convergence. They do not imply a general replacement of foreign-policy orientation, and they do not prove the full salience-to-vote mechanism. The outcome diagnostics instead point to selective convergence in UNGA voting, especially in issue areas where Brazil had room to move.

Future research can extend the design in three directions. First, cross-country media data would allow the salience step to be tested beyond Brazil. Second, qualitative work on policymakers and business actors could identify how the rank cue entered diplomatic deliberation. Third, similar rank-threshold designs could be applied to investment, aid, lending, or other economic ties where categorical milestones may become politically usable.

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Table 4: Brazil-China identical UNGA votes by year, 2005-2012.

Year	Resolutions	Identical votes	Identical votes (\%)	Mean similarity score
2005	97	80	82.5	0.892
2006	108	88	81.5	0.884
2007	93	78	83.9	0.903
2008	96	73	76.0	0.859
2009	83	72	86.7	0.934
2010	85	74	87.1	0.906
2011	96	83	86.5	0.927
2012	90	71	78.9	0.872

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8 Appendix

We present more details about the performance and checks on the fitted model and how we used ChatGPT to perform the NLP analysis.

8.1 Resolution-Level UNGA Vote Diagnostics

To make the Brazil-China movement in UNGA ideal points more concrete, Table 4 reports the annual share of resolutions in which Brazil and China cast identical votes from 2005 to 2012. The diagnostic uses roll-call votes from `unvotes` 0.3.0, based on Erik Voeten's UNGA voting data, and excludes missing or absent votes by either Brazil or China.

Note: Unit = roll-call resolutions with valid Brazil and China votes; mean similarity score equals 1 for identical votes, 0.5 for yes/abstain or no/abstain, and 0 for yes/no disagreement. Time window = 2005-2012 annual series. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

Figure 5 disaggregates the same diagnostic by substantive issue family. Because several areas already had high Brazil-China vote similarity before 2009, the figure should be read as descriptive evidence about where additional convergence was possible, not as a separate causal estimate.

Figure 5: Brazil-China identical UNGA votes by substantive issue area, 2005-2012. Points report annual percentages of roll-call resolutions in which Brazil and China cast the same vote; red lines are local linear fits estimated separately for 2005-2008 and 2009-2012. Note: Unit = percentage of roll-call resolutions with identical Brazil-China votes. Time window = 2005-2012 annual series; resolutions with multiple issue codes enter each corresponding substantive issue family. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

8.2 SDiD

8.2.1 Comparison of DiD, SCM, and SDiD estimating equations

To gain intuition for the SDiD estimator, it is useful to compare it with standard DiD and SCM. A DiD model estimates:

$$(\hat{\tau}^{\text{DiD}}_{ATT}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau_{ATT}, \mu, \alpha, \beta}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT}\right)^2$$

While SDiD assigns weights to the control group that are more similar to those of the treated unit and in time periods comparable to the pre-treatment period, classic DiD does not use any weights at all.

The SCM model can be written as follows:

$$(\hat{\tau}^{\text{SCM}}_{ATT}, \hat{\mu}, \hat{\beta}) = \underset{\tau_{ATT}, \mu, \beta}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \beta_t - D_{it}\tau_{ATT}\right)^2 \hat{w}^{\text{SCM}}$$

The equations show there is no unit fixed effect and no time weight in SCM. As a result, SCM cannot use unit-specific unobserved invariant characteristics to approximate the counterfactual outcome, and must consider the entire period when estimating the causal effect. SDiD combines unit weights (like SCM) with unit fixed effects and time weights (like DiD), addressing the limitations of both. The basic SDiD estimating equation is:

$$(\hat{\tau}^{\text{SDiD}}_{ATT}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau_{ATT}, \mu, \beta, \alpha}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau_{ATT}\right)^2 \hat{w}^{\text{SDiD}} \hat{\lambda}^{\text{SDiD}}_t$$

Note that SDiD retains unit fixed effects α_i (like DiD) while also assigning unit weights $\hat{w}^{\text{SDiD}}$ (like SCM) and time weights $\hat{\lambda}^{\text{SDiD}}_t$ (unique to SDiD).

8.2.2 SDiD weights and donor pool

Below we present the weights for the top 10 controls used in the main model.

We also fit a model for a donor pool of Latin American states (excluding Caribbean countries), which is more similar to Brazil than using countries worldwide. One reason to prefer a pool of Latin American countries is that a smaller donor pool helps avoid overfitting. When comparing results, the estimated causal effect is quite similar (a bit larger in absolute terms), which lends credibility to the main estimation. However, the small number of controls introduces noise and increases the uncertainty of the model. It is known that the usage of the placebo method to compute standard

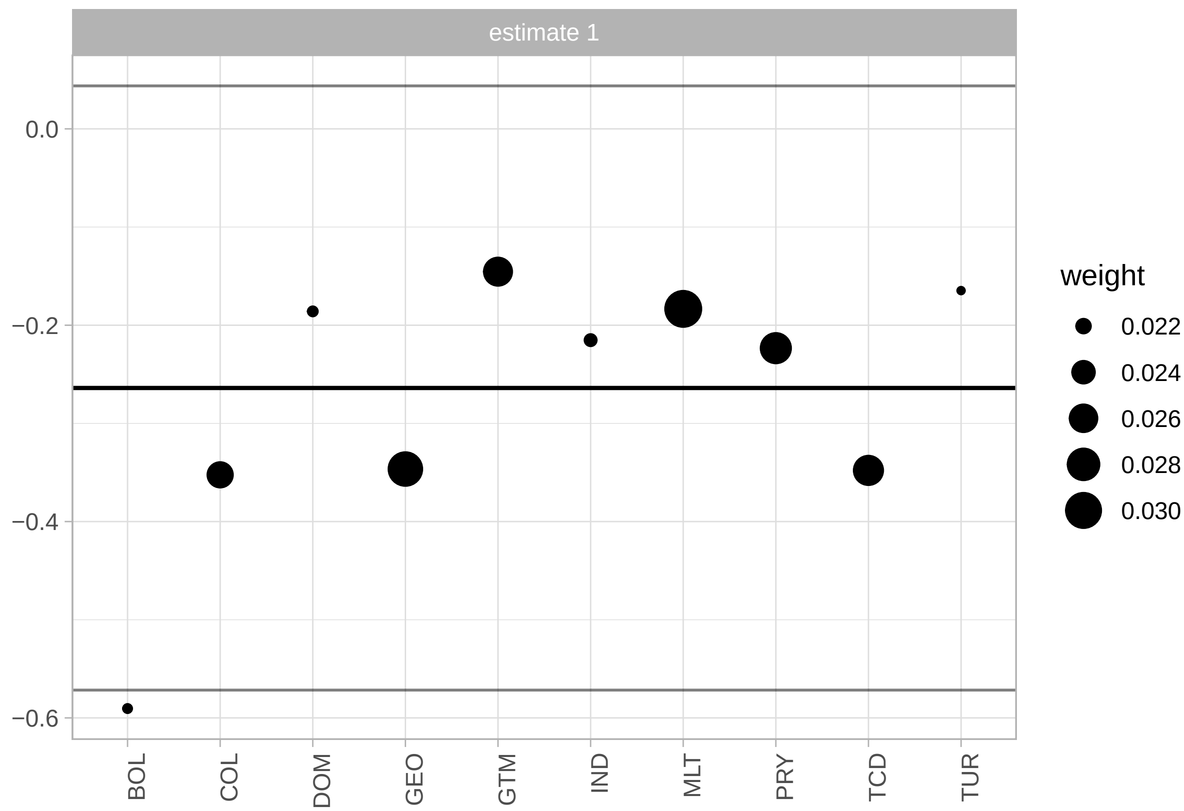

Figure 6: Weights of countries contributing to the synthetic control in the Brazilian SDiD model. Note: Unit = synthetic-control unit weights (unitless, sum to 1). Time window = weights are estimated from the pre-treatment period ending in 2008. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

errors inflates the uncertainty of the estimation. For this reason, we prefer the model with the larger donor pool. Nonetheless, it is reassuring that with a donor pool composed of Latin American countries, which are arguably more similar to Brazil, the estimate is quite similar. Below is the plot of the model fitted for Latin America.

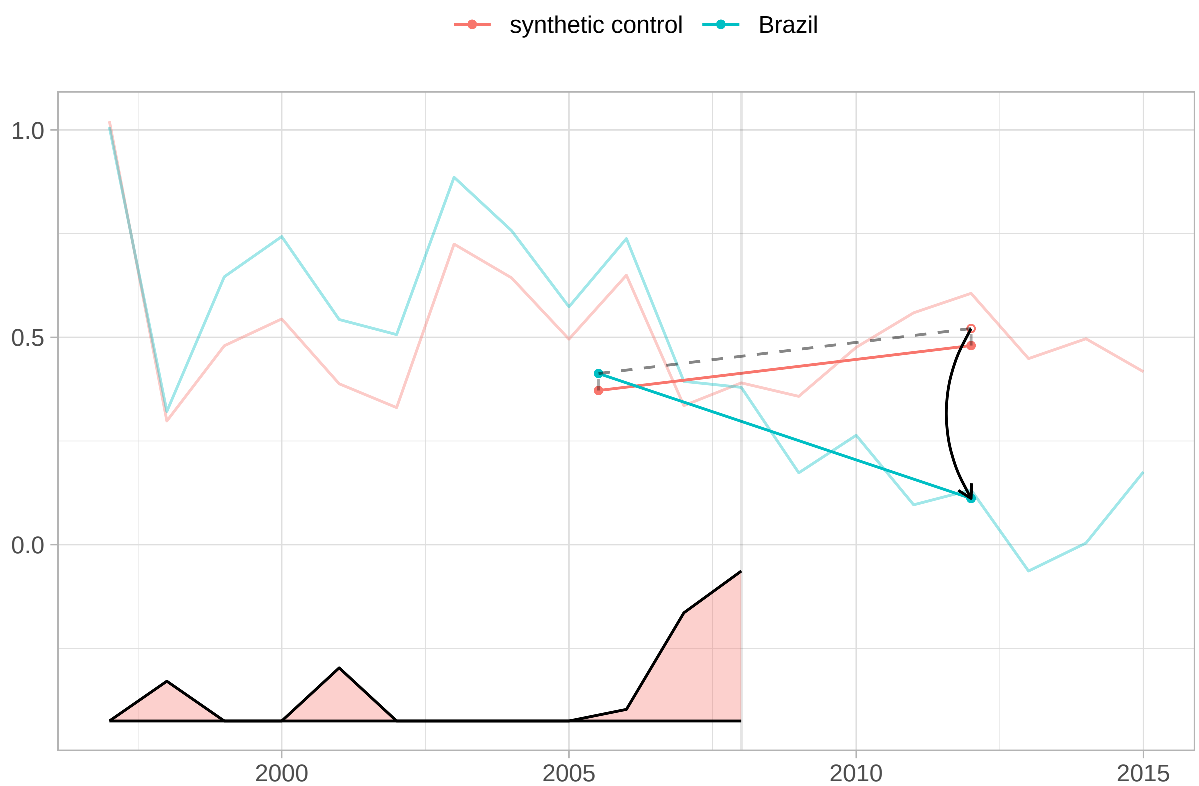

Figure 7: SDiD fit with a Latin American donor pool. Note: Unit = absolute UNGA ideal-point distance to China (points). Standard error = placebo-based (reported in text; not plotted here). Time window = pre-treatment and post-treatment years shown in the plot (treatment in 2009). Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

The estimate is -0.41 with a standard error of 0.2. In any case, it is interesting to examine the countries with the largest weights. Argentina, Guatemala, Paraguay, and Uruguay – all but Guatemala are founding members of Mercosur. They contribute the most to synthetic control.

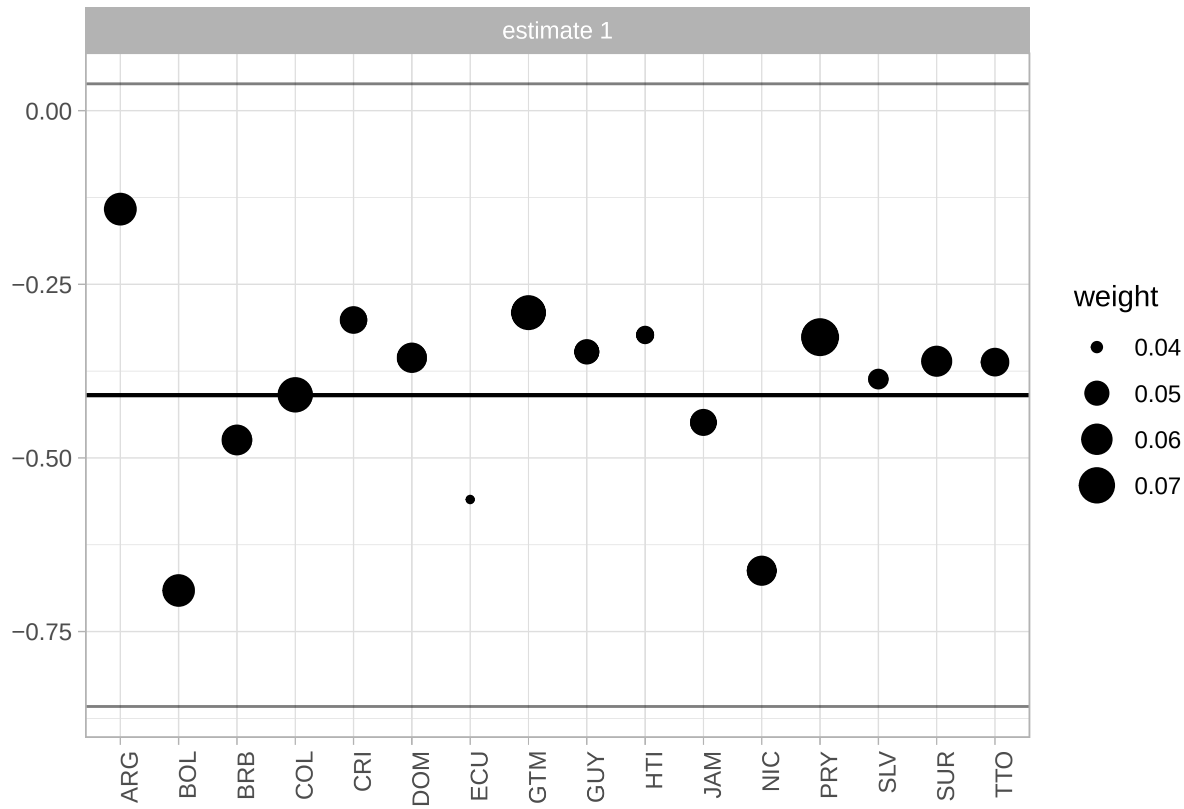

Figure 8: Synthetic-control donor weights for the Latin American SDiD specification. Note: Unit = synthetic-control unit weights (unitless, sum to 1). Time window = weights are estimated from the pre-treatment period ending in 2008. Treatment definition = Treatment indicator equals 1 from 2009 onward, when China becomes Brazil's top export destination displacing the USA, and 0 before 2009.

8.3 Cross-Country Robustness: Absorbing-Treatment Estimator (C&S)

As a diagnostic check, we also report an absorbing-treatment estimator using Callaway and Sant'Anna (2021). This specification is rebuilt from the same China-top-export-destination panel used by the main fect analysis. Treated units are countries for which China first becomes the number-one export destination in or after 2000 after an observed untreated year and then never loses the top position through the end of the panel. Countries with treatment reversals are excluded from this C&S comparison, not treated as controls. The baseline absorbing C&S estimate uses 11 treated units and 58 never-treated controls over 1990–2022; the covariate-adjusted specification uses 12 treated units and 51 controls over the complete-case window 1990–2020.

This check should be read as an absorbing-subset analogue, not as the same estimand as the switching-treatment fect model. The fect estimator uses entry and exit variation from the full switching panel, whereas C&S estimates the effect among persistent China-top cases only. The covariate-adjusted C&S result is additionally a complete-case sensitivity diagnostic: small treated groups and covariate adjustment can create overlap/convergence issues, and the changed panel window and treated/control counts mean that comparison with the baseline C&S column does not isolate covariate adjustment alone.

8.4 Cross-Country Diagnostic: No-Pretrend Equivalence Tests

Following Liu, Wang, and Xu (2024) and Chiu et al. (2025), we report equivalence tests for the no-pretrend assumption. Unlike the conventional placebo test (which tests H_0: no pre-trend), the equivalence test reverses the null to H_0: pre-trends are meaningfully large, providing positive evidence for the absence of pre-trends when rejected. The red dashed lines mark the equivalence range ($0.36\hat{\delta}_s$ or the user-specified threshold), while the grey dashed lines mark the minimum range.

8.5 ChatGPT Classification

Below is the coded instruction to the ChatGPT API to classify the subjects of the headlines.

```
system_block <- paste(
  'You are a Brazilian expert in International Relations and the political',
```

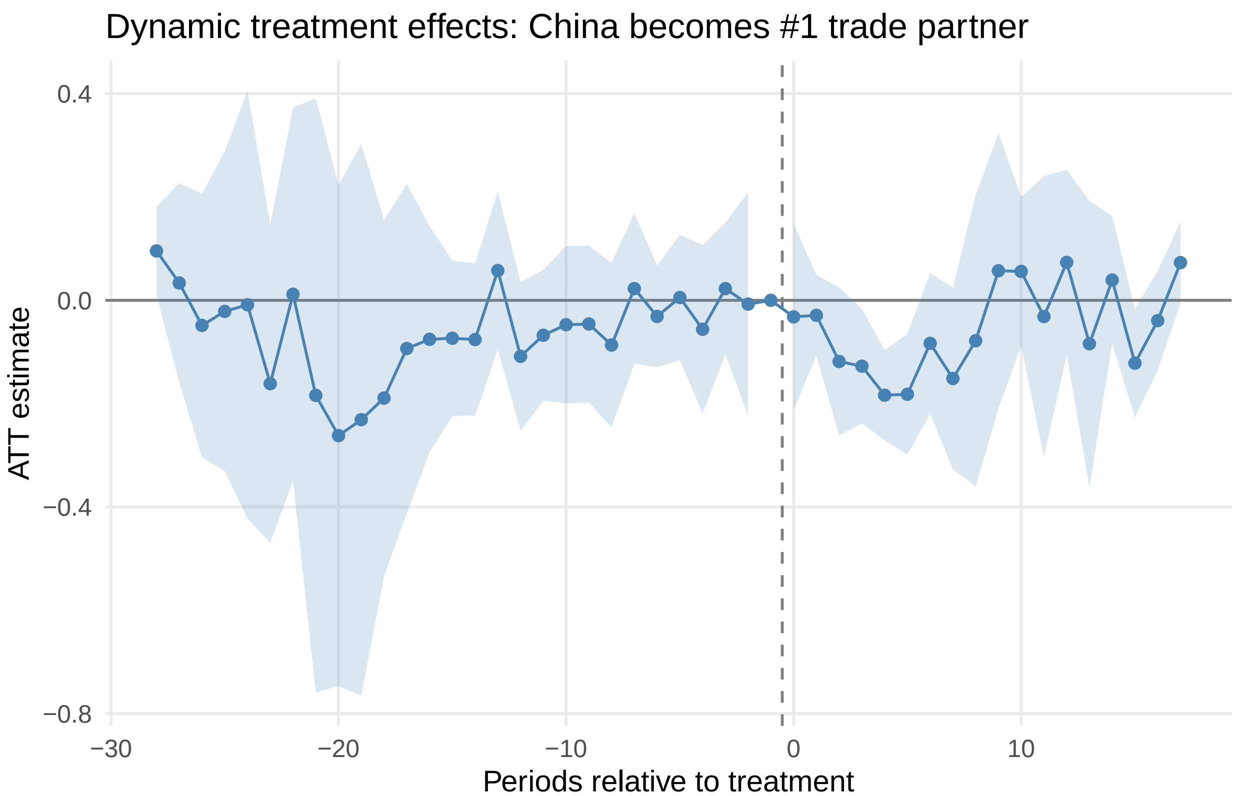

Figure 9: Dynamic treatment effects from the absorbing-treatment C&S specification. Note: Unit = ATT in absolute UNGA ideal-point distance to China (points). Standard error = analytical (95 percent confidence intervals shown). Cluster = country level. Time window = 1990-2022 balanced panel. Treatment definition = absorbing treatment: first observed year in or after 2000 in which China becomes the number-one export destination and never later loses that position.

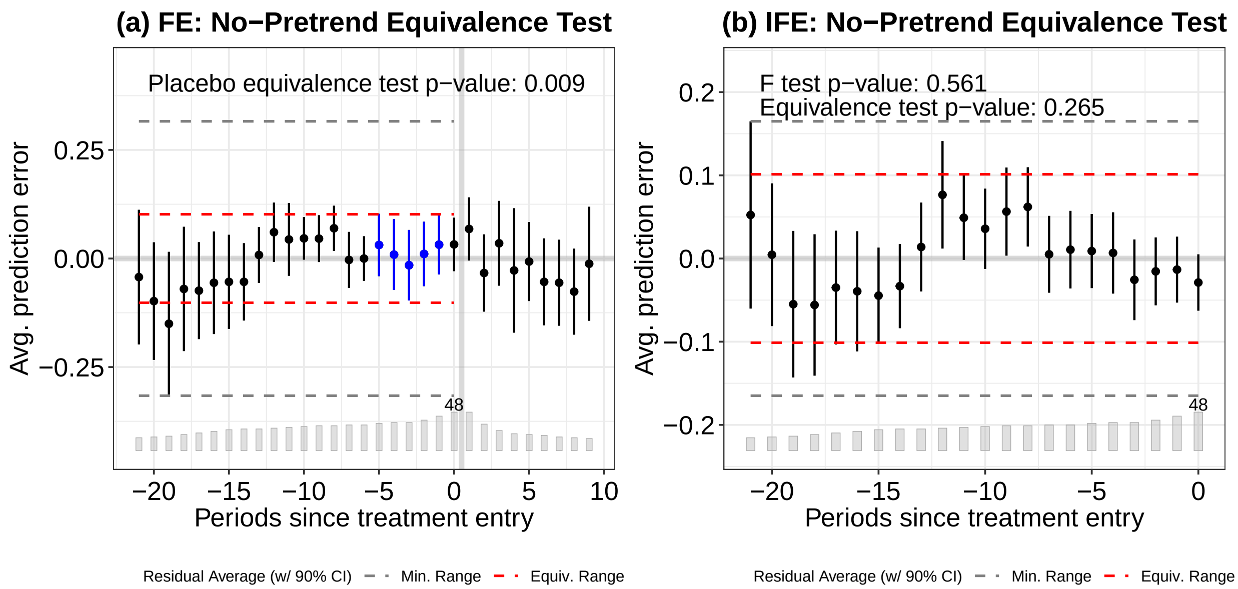

Figure 10: Equivalence tests for the no-pretrend assumption. (a) FE model: placebo equivalence test p = 0.009. (b) IFE model: F-test p = 0.561 and equivalence TOST p = 0.265. Note: Standard error = bootstrap (10,000 replications). Time window = 1990-2022 panel. Treatment definition = Treatment indicator equals 1 in any year China is currently the top export destination within the scope-conditioned panel; turns off if China loses the top position (switching treatment).

```
'economy of China-Brazil relations.  Classify the subject of EACH headline',
'when I say EACH, I mean all of them, even if repetitive or similar. Do classify **all
'using **one** label from this list exactly as written:',
'"diplomacy", "chinese economy", "china-brasil relations",',
'"disaster and accidents", "chinese politics and policy",',
'"sports, science, culture, other", "human rights",',
'"china-brazil trade", "health".\n\n',
# 13 static few-shot examples - leave them unchanged!
'Examples:\n',
'1. Presidente chinês visita o Brasil e participa de reunião com FHC -> china-brasil r
'2. Ministro chinês de Defesa faz visita oficial ao Brasil -> china-brasil relations\n
'3. Para governo, novo câmbio chinês só beneficia Brasil no médio prazo -> china-brazi
'4. China ultrapassa EUA como maior parceiro comercial do Brasil -> china-brazil trade
'5. China vai controlar preços de alimentos e combustíveis -> chinese economy\n',
```

```
  '6. Vendas de automóveis na China desaceleram em junho -> chinese economy\n',
  '7. China quer devolução das relíquias Yin, patrimônio da Unesco -> sports, science, c
  '8. Dinossauro encontrado na China esclarece origem das aves -> sports, science, cultu
  '9. OMS confirma mais de 4.600 casos de gripe suína; China registra 2º caso -> health\
  '10. Marinha chinesa busca "cansar" barcos japoneses em águas disputadas -> diplomacy\
  '11. Acidente após casamento deixa 16 mortos na China -> disaster and accidents\n',
  '12. Jornal chinês sai das bancas por foto da praça Tiananmen -> human rights\n',
  '13. China é parceira estratégica contra crise econômica, diz UE -> chinese economy',
  'return as in the examples, a string with the headline and the classification'
)
```

It is important to highlight that the process was not streamlined. We faced several problems, such as ChatGPT refusing to classify subjects (saying they were repetitive and too similar), changing the original headline, which made matching back to the original dataset difficult, and other similar problems. As a result, we had to rerun the instructions a few times on a sample and tweak it with some prompt engineering until arriving at that final version. As a result, this is the only part of the paper not fully reproducible, in the sense that another request for classification, with the same instructions, would yield a few different categorizations and a few different original headlines changed, along with some refusals to classify texts.

Below we provide a random sample of the headlines and the classification provided by ChatGPT.

TABLE 1 — Examples of Classified Headlines

Random sample of 20 headlines with ChatGPT-assigned topic

Headline	Date
china-brazil relations	
Disputa com Brasil é 'conflito menor' para China, dizem analistas	2005-10-05
Lula vai se encontrar com presidente da China na segunda	2004-05-23
china-brazil trade	
China lidera as aquisições de empresas brasileiras	2013-03-01
China aceita nova regra brasileira e suspende embargo à soja	2004-06-21
chinese economy	
PIB da China dobra participação no total mundial em 5 anos	2011-03-25
BHP prevê produção recorde de minério de ferro e aposta na China	2012-01-18
chinese politics and policy	
China ameaça Google com "consequências" caso empresa largue autocensura	2010-03-12
China anuncia fechamento de 700 websites	2004-08-01
diplomacy	
China não quer interferência dos EUA na questão do Tibete	2011-09-28
Chega a Taiwan primeiro vôo direto da China em quase 60 anos	2008-07-04
disaster and accidents	
Incêndio em prédio residencial deixa 12 mortos na China	2010-07-19
China detona barragens e prédios danificados; veja vídeo	2008-06-06
health	
Uma em três chinesas da zona rural se mata por ano	2002-11-29
China suspende viagens ao Tibete para controlar Sars	2003-05-13
human rights	

Figure 11: Random sample of China-related Folha de Sao Paulo headlines and ChatGPT classifications. Note: Unit = headline text and category labels. Time window = sample drawn from the 2000-2014 corpus.

Table 5: Validation of ChatGPT classification against manual coding (N = 100). Overall accuracy: 88.0 percent.

Category	N	Correct	Accuracy (%)
china-brazil trade	6	6	100.0
disaster and accidents	13	13	100.0
health	6	6	100.0
chinese economy	25	24	96.0
human rights	8	7	87.5
diplomacy	12	10	83.3
sports, science, culture, other	17	14	82.4
china-brazil relations	5	4	80.0
chinese politics and policy	8	4	50.0
Overall	100	88	88.0

8.5.1 Validation of ChatGPT Classification

To assess the reliability of the automated classification, one of the authors independently coded a stratified random sample of 100 headlines (with a minimum of 5 per category). Table 5 reports the agreement rate between the ChatGPT labels and the manual coding.

Note: Unit = headline-level counts and accuracy (percent) by category. Time window = stratified validation sample from the 2000-2014 headline corpus.

The overall accuracy is 88%. The lowest agreement is in the “chinese politics and policy” category (50%), where several headlines were manually reclassified as “diplomacy”. The categories most relevant to our analysis — “china-brazil trade” and “disaster and accidents” — achieved 100% agreement.

8.6 Cross-Country: Treated Countries and Treatment-Switching Diagnostic

Table 6 describes the 32 treated countries where China becomes the top export destination in the scope-conditioned panel. For each country, we report the treatment year, whether the treatment is absorbing (China remains #1) or switching (China later loses the top position), and the mean UNGA voting distance to China before and after treatment.

Table 6: Descriptive statistics of treated countries where China becomes the top export destination in the scope-conditioned cross-country panel.

Country	ISO3c	Treatment year	Absorbing	Switches	Mean dist. (pre)	Mean dist. (post)	Diff
Angola	AGO	2007	Yes	1	0.513	0.247	-0.266
Armenia	ARM	2020	No	2	1.062	0.780	-0.281
Benin	BEN	2004	No	6	0.507	0.289	-0.218
Brazil	BRA	2009	Yes	1	0.733	0.403	-0.330
Chile	CHL	2008	Yes	1	0.800	0.433	-0.367
Congo - Kinshasa	COD	2008	Yes	1	0.553	0.255	-0.298
Congo - Brazzaville	COG	2003	No	3	0.571	0.219	-0.352
Ethiopia	ETH	2009	No	6	0.542	0.221	-0.320
Gabon	GAB	2017	Yes	1	0.339	0.191	-0.148
Ghana	GHA	2019	No	2	0.312	0.290	-0.023
Guinea	GIN	2019	No	3	0.320	0.383	0.062
Equatorial Guinea	GNQ	2006	No	5	0.674	0.357	-0.317
Indonesia	IDN	2013	No	3	0.324	0.487	0.164
Iraq	IRQ	2014	No	3	0.359	0.422	0.063
Japan	JPN	2010	No	6	1.736	1.303	-0.432
South Korea	KOR	2003	Yes	1	1.536	1.420	-0.117
Kuwait	KWT	2018	Yes	1	0.440	0.343	-0.096
Liberia	LBR	2012	No	4	0.498	0.649	0.151
Myanmar (Burma)	MMR	2014	Yes	1	0.450	0.390	-0.060
Malaysia	MYS	2009	Yes	1	0.395	0.258	-0.137
Peru	PER	2012	No	3	0.758	0.397	-0.362
Philippines	PHL	2005	Yes	1	0.486	0.272	-0.214
North Korea	PRK	2010	No	3	0.700	0.630	-0.070
Saudi Arabia	SAU	2015	Yes	1	0.386	0.241	-0.145
Singapore	SGP	2013	No	4	0.519	0.288	-0.232
Sierra Leone	SLE	2012	Yes	1	0.391	0.241	-0.150
Chad	TCD	2019	No	2	0.349	0.337	-0.013
Thailand	THA	2009	No	2	0.461	0.244	-0.217
Uruguay	URY	2013	Yes	1	0.825	0.288	-0.538
Venezuela	VEN	2021	Yes	1	0.674	1.493	0.819
Vietnam	VNM	2018	No	2	0.548	0.362	-0.186
South Africa	ZAF	2010	Yes	1	0.602	0.247	-0.355

Note:
Absorbing = China remains the top export destination from first treatment entry through the end of the sample. Switching = China later loses the top position. Mean dist. = absolute distance to China in UNGA ideal points. Diff = post minus pre.

8.7 Leave-One-Out Sensitivity Analysis

Table 7 reports the fect IFE ATT when each treated country is excluded from the sample. This check assesses whether the cross-country result is driven by any single influential observation.

Table 7: Leave-one-out sensitivity: fect IFE ATT when each treated country is excluded.

	Dropped country	ATT	SE	p-value	r^*
r	Full model	-0.060	0.032	0.065	1
r32	Angola	-0.055	0.033	0.094	1
r1	Armenia	-0.061	0.032	0.058	1
r2	Benin	-0.060	0.033	0.067	1
r3	Brazil	-0.059	0.033	0.079	1
r4	Chile	-0.061	0.034	0.070	1
r5	Congo - Kinshasa	-0.054	0.042	0.199	2
r6	Congo - Brazzaville	-0.056	0.053	0.295	2
r7	Ethiopia	-0.056	0.051	0.267	2
r8	Gabon	-0.054	0.050	0.276	2
r9	Ghana	-0.058	0.048	0.233	2
r10	Guinea	-0.057	0.051	0.263	2
r11	Equatorial Guinea	-0.058	0.052	0.265	2
r12	Indonesia	-0.058	0.032	0.069	1
r13	Iraq	-0.051	0.052	0.319	2
r14	Japan	-0.060	0.033	0.065	1
r15	South Korea	-0.081	0.026	0.002	1
r16	Kuwait	-0.059	0.033	0.073	1
r17	Liberia	-0.052	0.051	0.312	2
r18	Myanmar (Burma)	-0.051	0.030	0.087	1
r19	Malaysia	-0.055	0.032	0.088	1
r20	Peru	-0.057	0.033	0.085	1
r21	Philippines	-0.060	0.033	0.073	1
r22	North Korea	-0.064	0.034	0.064	1
r23	Saudi Arabia	-0.058	0.033	0.074	1
r24	Singapore	-0.060	0.033	0.064	1
r25	Sierra Leone	-0.061	0.033	0.064	1
r26	Chad	-0.060	0.033	0.068	1
r27	Thailand	-0.058	0.033	0.078	1
r28	Uruguay	-0.055	0.033	0.099	1
r29	Venezuela	-0.051	0.030	0.089	1
r30	Vietnam	-0.068	0.034	0.043	1
r31	South Africa	-0.059	0.034	0.077	1